Nikolas Espinoza

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Education

University of California, San Diego — La Jolla, CA

B.S. in Computer Science — *Sep 2025 – June 2029*

- **Coursework:** Data Structures and Object-Oriented Design, Systems Programming, Discrete Mathematics and Graph Theory, Mathematical Reasoning, Intro to MATLAB

Experience

San Diego Supercomputer Center — La Jolla, CA

Research Data Services Intern — *September 2026 – June 2027*

- Selected for a **return offer** to develop and ship features across multiple **Research Data Services** projects through Spring 2027

*Project Developer Intern – **Cook Out*** — *July 2026 – September 2026*

- Developed **Cook Out**, a full-stack social cooking iOS platform, in a **6-member** team using **TypeScript** and **Expo/React Native**, coordinating through **GitLab and Slack**
- Designed a normalized **PostgreSQL** schema (8+ tables) with **Prisma ORM**, automated FK constraints, triggers, and cursor-based pagination to optimize social feed queries
- Implemented secure authentication with **JWT/Passport.js** and built transactional email pipelines by integrating **Cloudflare** with **Resend**

Projects

Tweetify | *MongoDB, Express, React, Node.js* | ACM Projects — Jan 2026 – Mar 2026

- Co-developed a dynamic **MERN-stack media platform** combining Twitter and Spotify features alongside a 4-person team
- Implemented scalable **RESTful API endpoints** and **Mongoose** schemas to dynamically retrieve and stream user audio playlists and social feed data

Point-A-Sketch | *Python, OpenCV, Adafruit M0 Express* | SIPP Hackathon — July 2025

- Engineered an **embedded art program** on an **Adafruit M0 Express board**, interpreting hand gestures and acoustic sensor inputs into real-time canvas controls
- Trained an **MLP regression model** using **OpenCV** for real-time gesture recognition, leveraging pixel-level inference to accurately interpret drawing direction

Activities and Leadership

ACM Hack Event Director — *November 2025 – Present*

- Planned and delivered technical workshops alongside fellow interns covering **Java, React, and HTML/CSS** fundamentals and related libraries for students across varying levels of experience

SkillsUSA Lead – Video Game Development — *September 2021 – June 2025*

- Annually led 2–4 member teams in developing original video games using the **Unity Game Engine**
- Coordinated weekly team meetings and maintained structured deadlines to ensure consistent progress across multiple development stages

Technical Skills

Languages: C#, Java, Python, MATLAB, JS, TypeScript, C, SQL, Lua

Frameworks: React, React Native, Expo, Node.js, Express, Vite

Databases: PostgreSQL, MongoDB

Developer Tools: Git, GitLab, Unity, pgAdmin, Cloudflare

Libraries: Prisma, Mongoose, Passport.js, JWT, bcrypt, Resend, MaterialUI, NumPy, OpenCV, PySerial, Unity NetCode